Dynamic Financial Simulation

Monte Carlo Pricing of Path-Dependent Asian Options and Protective-Put Hedging Analysis

Monte Carlo Simulation • Dynamic Systems • Derivative Valuation • Financial Risk Management

Portfolio Project Report

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Abstract

This project develops a Monte Carlo framework for two connected **financial decision problems**: **valuation of path-dependent Asian options** and **evaluation of a protective-put hedging strategy**. The model represents stock-price evolution as a dynamic stochastic system in which each simulation replication generates a possible market trajectory.

For the **Asian-option** module, weekly stock prices are simulated over a one-year horizon using a geometric random walk, and the arithmetic average price along each path determines discounted call and put payoffs. One thousand simulated paths are used to estimate expected payoffs, confidence intervals, payoff probabilities, and distributional shape. **The second module** simulates 2,000 six-month terminal stock-price outcomes and compares an unhedged stock position with a stock-plus-put strategy. The analysis emphasizes not only expected payoff but also variance, downside probability, and worst-case simulated loss.

Results show strong asymmetry between the simulated Asian call and put payoff distributions under the specified assumptions. In the hedging experiment, the protective put materially compresses payoff variance and limits severe losses, even though the probability of a small negative payoff rises because the option premium shifts the payoff distribution downward. Convergence diagnostics show that running Monte Carlo means become progressively more stable as the number of replications increases. The project illustrates how simulation can connect stochastic financial modeling, derivative valuation, and risk-management decisions within a single analytical workflow.

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1. Problem Background and Analytical Motivation

Financial decisions are inherently intertemporal: assets and liabilities are allocated today while future market conditions remain uncertain. A central modeling challenge is that market states do not merely vary across scenarios; they evolve dynamically and stochastically through time. A stock price observed next week depends on its current state and on a new random innovation, and the sequence of these linked states forms a possible trajectory of the market. This distinction matters whenever a financial payoff depends on the path taken by the underlying asset rather than only on its final value.

Derivatives provide a natural setting for this problem. A derivative derives its value from an underlying asset, index, rate, or related state variable. Options give the holder a right rather than an obligation: a call provides the right to buy, while a put provides the right to sell at a specified strike price. **European options** depend on the underlying price at maturity. **Asian options** differ because their payoff is linked to an average asset price over time, making them path dependent. Consequently, valuation requires information about the full simulated trajectory rather than a single terminal draw.

The project therefore addresses two related questions. **First, how can Monte Carlo simulation be used to characterize and value path-dependent Asian option payoffs when stock prices evolve dynamically? Second, how does a put option change the risk-return profile of a stock position when it is used as insurance against adverse price movements?** These questions connect valuation and risk management: the same derivative that has a simulated value can also be used to reshape the distribution of portfolio outcomes.

2. Theoretical Framework

2.1 Dynamic systems and sample paths

Let X_t denote the state of a financial system in period t. The process X = {X_0, X_1, ..., X_T} describes the system's evolution, while one realization (x_0, x_1, ..., x_T) is a sample path. A dynamic simulation requires an initial condition and a recursion linking consecutive states. In this project, the initial state is the stock price S_0 and the recursion is a geometric random walk.

$$S_t = S_(t\text{-}1)\ \exp[(\mu - \sigma^2/2)\Delta t + \sigma\sqrt{(\Delta t)}\ Z_t], \quad Z_t \sim N(0,1)$$

Here μ is the drift, σ is volatility, Δt is the time step, and Z_t is an independent standard-normal innovation. The exponential specification keeps simulated stock prices positive and compounds shocks through time. For the one-year Asian-option experiment, Δt = 1/52, so each replication produces a weekly trajectory. With N replications, the simulated sample is naturally viewed as a collection of N paths, each containing the sequence of weekly states.

2.2 Option payoffs and path dependence

For a European option with maturity T, strike K, risk-free rate r, and terminal stock price S_T, the discounted payoff depends only on S_T. The corresponding call and put payoff functions are:

$$\text{Call payoff} = \max(S_T - K, 0)\ \exp(-rT)$$

$$\text{Put payoff} = \max(K - S_T, 0)\ \exp(-rT)$$

An average-price Asian option replaces the terminal stock price with the arithmetic average of prices along the path. Let S_avg denote the average simulated stock price over the monitoring dates. The workbook computes discounted Asian call and put payoffs as:

$$C_R0 = \max(S_avg - K, 0)\ \exp(-rT)$$

$$P_R0 = \max(K - S_avg, 0)\ \exp(-rT)$$

This path dependence is the reason the dynamic simulation is essential. Two paths may finish at similar terminal prices yet produce different average prices and therefore different Asian-option payoffs.

2.3 Monte Carlo estimation and simulation uncertainty

Monte Carlo simulation approximates quantities of interest by repeatedly generating random outcomes from the specified model and averaging the resulting payoffs. The expected discounted payoff is estimated by the sample mean across replications. The workbook also reports 95% confidence intervals using the simulated standard deviation and the standard error of the mean. Because the model uses random draws, any single workbook recalculation can produce slightly different estimates. The appropriate interpretation is therefore distributional and probabilistic rather than deterministic.

Monte Carlo methods are widely used in computational finance for derivative valuation and path-dependent claims. Academic finance references similarly organize simulation around stochastic state generation, discounted payoffs, sampling error, and variance or convergence diagnostics. MIT's Analytics of Finance materials, for example, treat stochastic calculus, option pricing, and simulation methods as linked components of computational asset pricing, while classic references by Boyle, Broadie, and Glasserman and by Glasserman develop Monte Carlo methods for security pricing and financial engineering.

3. Simulation Design and Implementation

3.1 Asian option module

Parameter	Workbook value
Initial stock price, S0	100
Drift, μ	0.08
Volatility, σ	0.12
Strike price, K	97
Expiration, T	1 year
Step size, Δt	1/52 year
Risk-free rate, r	0.059
Monte Carlo paths, N	1,000

Each replication begins at S0 = 100 and generates weekly stock prices recursively for one year. The arithmetic average price is calculated from the simulated path. That average is then translated into discounted Asian call and put payoffs. Repeating the process 1,000 times creates empirical payoff distributions from which the workbook calculates expected payoffs, confidence intervals, threshold probabilities, and histograms.

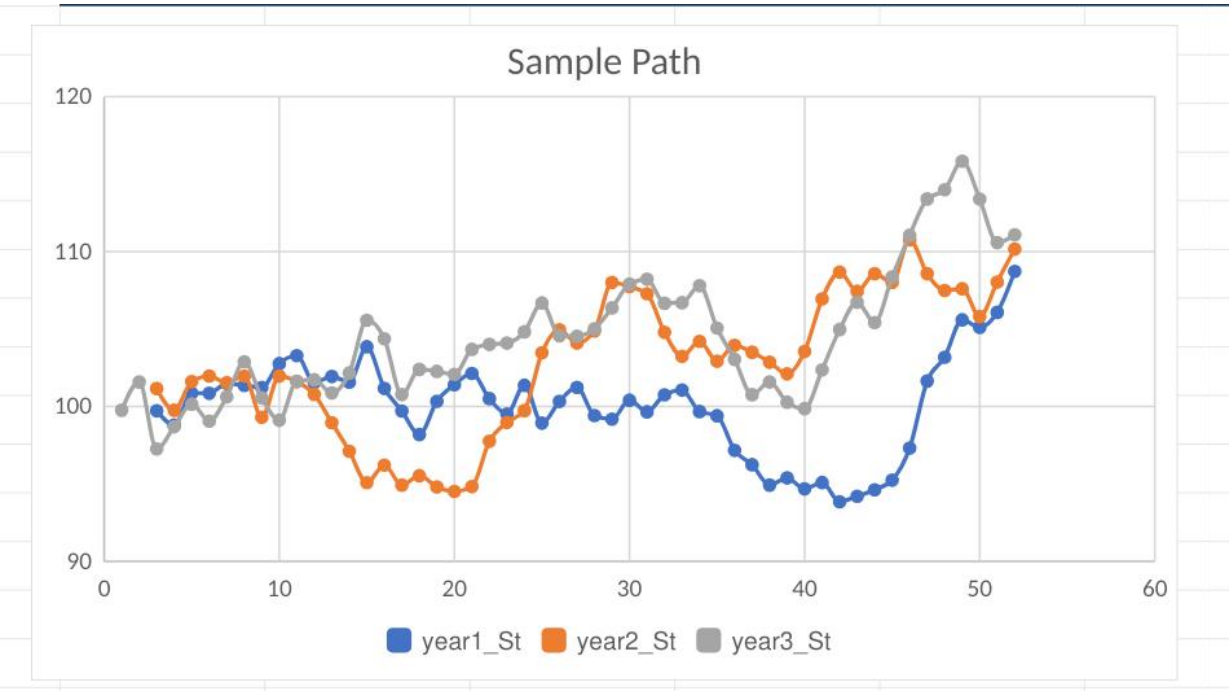


Figure 1. Example simulated weekly stock-price paths from the Asian-option model. Each line represents one possible trajectory generated by the dynamic recursion.

3.2 Hedging module

Parameter	Workbook value
Initial stock price, S0	100
Drift, μ	0.08
Volatility, σ	0.12
Put strike, K	97
Expiration, T	0.5 year
Step size, Δt	0.5 year
Risk-free rate, r	0.059
Monte Carlo replications, N	2,000
Observed put premium, c	0.57

The hedging experiment uses one six-month transition per replication to generate S_T. The discounted European put payoff is R0 = max(K − S_T, 0)exp(−rT). Two investment outcomes are then evaluated under the same simulated stock scenarios. Strategy 1 holds the stock without protection. Strategy 2 combines the stock with the put and subtracts the option premium. Because both strategies are evaluated against identical simulated states, differences in their payoff distributions isolate the effect of the hedge.

The comparison deliberately uses multiple risk measures. Expected payoff describes average simulated performance; variance measures dispersion; downside-risk indicators estimate the probability of a negative payoff; and simulated minima and maxima reveal the severity of extreme outcomes observed in the experiment. This prevents the hedge from being judged by a single metric.

4. Results

4.1 Asian option valuation and payoff distributions

Instrument	Mean discounted payoff	95% CI	P(payoff > 1.1)	95% CI for probability
Asian put	0.4651	[0.3716, 0.5585]	0.1150	[0.0951, 0.1349]
Asian call	7.9093	[7.5089, 8.3098]	0.8160	[0.7920, 0.8400]

Under the workbook's specified assumptions, the simulated Asian put has an expected discounted payoff of 0.4651, with a 95% confidence interval from 0.3716 to 0.5585. The corresponding probability that the put payoff exceeds 1.1 is 11.50%, with a 95% interval of approximately 9.51% to 13.49%. The Asian call produces a much larger expected discounted payoff of 7.9093, with a 95% confidence interval from 7.5089 to 8.3098. Its probability of exceeding 1.1 is 81.60%, with a 95% interval from approximately 79.20% to 84.00%.

The payoff histograms clarify why the two means differ so substantially. The Asian put distribution contains a large point mass at zero: 85.0% of simulated put payoffs fall in the zero bin. By contrast, the call distribution is spread across a wider range of positive payoffs, with only 15.0% in the zero bin. This asymmetry is consistent with the model inputs: the initial stock price is 100, the strike is 97, and the assumed drift is positive. Under these assumptions, many simulated average prices remain above the strike, leaving the put out of the money while supporting positive call payoffs.

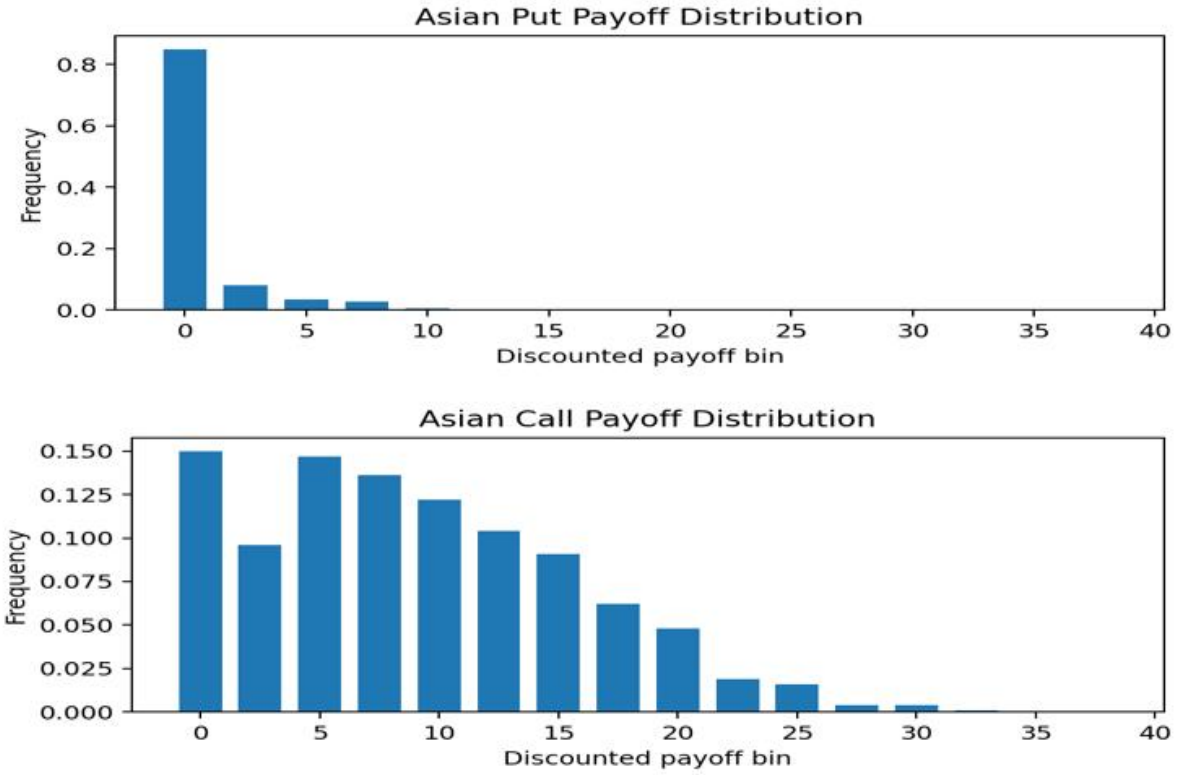


Figure 2. Workbook histograms for discounted Asian put and call payoffs. The put distribution is concentrated at zero, while the call distribution is more broadly distributed over positive payoff levels.

4.2 Terminal stock price and put valuation in the hedging experiment

Across 2,000 simulated six-month scenarios, the mean terminal stock price is 104.0656. The 95% confidence interval for the mean is [103.6781, 104.4531], and the simulated probability that the terminal stock price is below the initial price is 33.45%. The observed terminal-price range in this simulation is approximately 75.4938 to 140.0780.

The simulated discounted European put payoff has a mean of 0.9674 and a 95% confidence interval of [0.8594, 1.0754]. The workbook's quoted put premium is 0.57, which lies below this simulation interval. Within the classroom model and the specific stochastic assumptions used here, the quoted premium is therefore lower than the Monte Carlo estimate of the expected discounted payoff. This comparison should be interpreted as a model-based simulation result rather than as a claim of an arbitrage-free market price, because the stock process in the workbook uses the specified drift $\mu = 0.08$ rather than replacing the drift with the risk-free rate under a risk-neutral pricing measure.

4.3 Hedging strategy comparison

Metric	Stock only	Stock + put
Mean discounted payoff	3.9474	4.3448
Payoff variance	73.6949	52.4710
Probability of negative payoff	33.45%	36.50%
Maximum simulated payoff	38.9130	38.3430
Minimum simulated payoff	−23.7938	−3.4828

The protective put changes the shape of the payoff distribution rather than improving every risk statistic simultaneously. The stock-plus-put strategy has a slightly higher simulated mean payoff in this run, 4.3448 versus 3.9474, while its variance falls from 73.6949 to 52.4710, a reduction of approximately 28.8%. The simulated minimum payoff also improves from −23.7938 without hedging to −3.4828 with the put, showing that the option substantially limits the severity of extreme losses. These improvements, however, are not sufficient to determine the project's final hedging decision because Q3 also evaluates the downside-risk probability and the cost of the hedge.

The **decisive Q3 result** is that the probability of a negative payoff rises from 33.45% for the stock-only position to 36.50% for the stock-plus-put position. Purchasing the option requires the 0.57 premium, which shifts the hedged payoff distribution downward in states where the protection is not sufficiently valuable. The hedge can therefore create more small negative outcomes while eliminating much larger losses. Under the decision criterion implemented in the workbook, the simulation therefore supports **No Hedging**. This conclusion does not mean that the put provides no protection; rather, the protection against loss severity is obtained at a cost and does not improve the downside-risk probability used for the final decision.

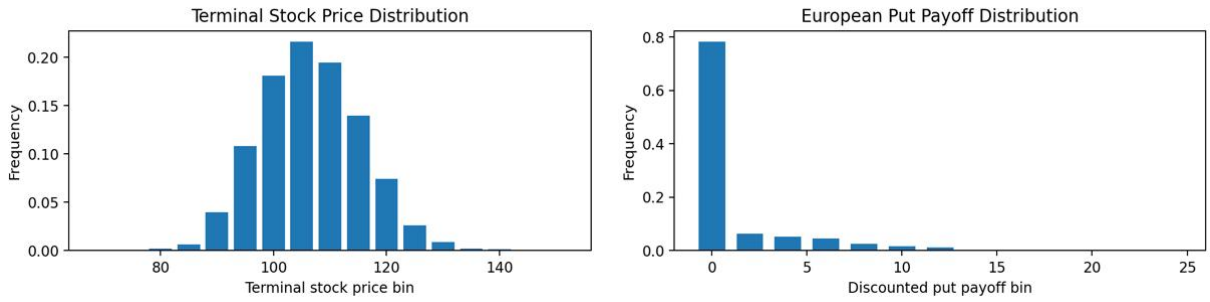


Figure 3. Workbook distribution views for the terminal stock price and put payoff in the hedging simulation. The empirical distributions summarize the 2,000 simulated market states used in the strategy comparison.

5. Monte Carlo Convergence and Reliability

Because Monte Carlo estimates are sample averages, early estimates can move substantially when only a small number of replications have been observed. The workbook therefore tracks running means at regular checkpoints. For the Asian option, the put running mean moves from 0.035 at 25 paths to 0.389 at 50 paths and then fluctuates within a much narrower region as the simulation approaches 1,000 paths, finishing at 0.4651. The call estimate begins at 10.345 for 25 paths, falls toward the high-7 to low-8 range, and finishes at 7.9093.

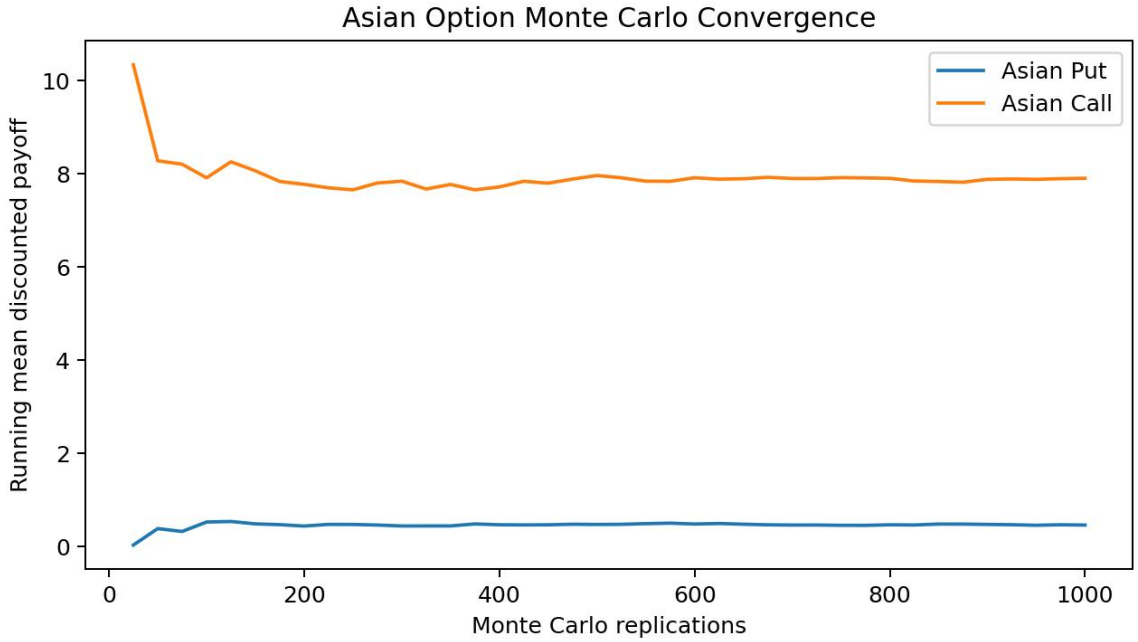


Figure 4. Running Monte Carlo mean payoffs for the Asian put and call. Estimates fluctuate more at small sample sizes and become progressively more stable as the number of simulated paths increases.

The hedging estimates show a similar pattern. At 50 replications, the running mean payoff is 2.978 for stock only and 2.920 for stock plus put. By 1,000 replications the corresponding estimates are 4.131 and 4.499; by 2,000 replications they settle at 3.9474 and 4.3448. The curves do not become perfectly flat—Monte Carlo error remains finite—but the incremental effect of additional replications generally becomes smaller. The convergence analysis therefore provides a useful diagnostic that the reported results are not driven solely by a very small set of early random draws.

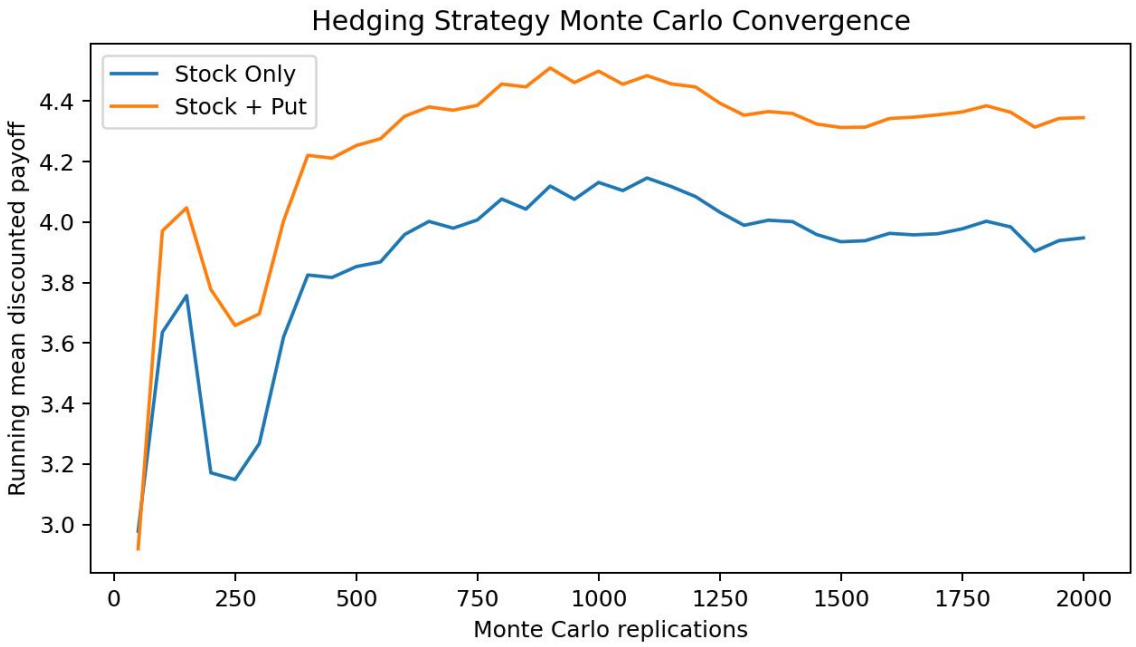


Figure 5. Running mean payoffs for the unhedged and protective-put strategies across the 2,000 simulated scenarios.

6. Interpretation and Managerial / Analytical Implications

6.1 Path dependence changes what must be simulated

The Asian-option exercise demonstrates a broader modeling principle: when an outcome depends on the history of a system, simulating only an end state discards economically relevant information. The average-price payoff requires the entire weekly trajectory. Similar logic appears in operational and business settings involving cumulative demand, inventory trajectories, customer histories, or exposure accumulated through time. The analytical lesson is therefore broader than derivative pricing: the simulation unit must match the structure of the decision problem.

6.2 Distributional analysis is more informative than a point estimate

A single expected payoff cannot describe the asymmetry, zero-payoff mass, or tail behavior of an option. The histograms, confidence intervals, and payoff probabilities show materially different risk profiles for the Asian call and put even though both are generated from the same underlying stock process. For decision makers, this supports reporting a distribution of possible outcomes alongside the mean rather than treating the Monte Carlo average as a complete description of uncertainty.

6.3 Hedging should be evaluated by the risk objective

The protective-put results illustrate why risk management requires an explicit decision criterion. In this project, Q3 uses the simulated risk comparison to determine whether the position should be hedged. The probability of a negative payoff increases from 33.45% without hedging to 36.50% with the put, while the hedge also requires a 0.57 premium. On that basis, the workbook's decision is No Hedging. The simulation nevertheless shows why this decision is not equivalent to saying that the put has no risk-management effect: payoff variance falls by about 28.8% and the minimum simulated payoff improves by more than 20 payoff units. Thus, the put reduces the severity and dispersion of losses, but it does not improve the downside-risk probability that drives the project's final decision.

6.4 The option premium is the cost of insurance

The put premium creates an explicit trade-off. In favorable or moderately unfavorable states, the premium can reduce the hedged strategy's payoff relative to holding the stock alone; in sufficiently adverse states, the put becomes valuable and creates a floor-like effect. For the present decision problem, the additional protection does not compensate for the higher simulated probability of a negative payoff under the criterion used in Q3, so the recommended model decision remains No Hedging. More generally, this illustrates that insurance should be evaluated against the specific risk measure and decision objective rather than assumed to be preferable simply because it truncates extreme losses.

7. Limitations

The results are conditional on the workbook's assumptions and should not be interpreted as forecasts of a particular traded security. The stock process uses constant drift and volatility, independent standard-normal shocks, and a geometric random-walk specification. Real markets can exhibit time-varying volatility, jumps, serial dependence, liquidity effects, transaction costs, and parameter uncertainty. The Asian option uses discrete weekly monitoring, so its average depends on the selected monitoring frequency.

The pricing interpretation also requires care. The workbook simulates the stock process using $\mu = 0.08$ while discounting payoffs at $r = 0.059$. Standard arbitrage-free derivative pricing typically evaluates discounted payoffs under a risk-neutral probability measure in which expected asset returns are aligned with the risk-free rate. The present results should therefore be described as Monte Carlo expected discounted payoffs under the workbook's specified model assumptions. This distinction preserves the educational purpose of the model while avoiding a stronger theoretical pricing claim than the implemented process supports.

Finally, Monte Carlo estimates retain sampling error. Confidence intervals quantify part of this uncertainty, and the running-mean diagnostics show increasing stability, but another recalculation will produce somewhat different numerical results because the simulation is driven by random draws. Larger samples, variance-reduction techniques, parameter sensitivity analysis, and repeated independent simulation batches could further strengthen numerical reliability.

8. Conclusion

This project integrates **dynamic stochastic modeling**, **Monte Carlo simulation**, **derivative payoff analysis**, and **financial risk management**. The **Asian-option** module shows how a geometric random walk can generate complete stock-price paths and how path averages can be translated into discounted option payoffs. The resulting empirical distributions reveal substantial asymmetry between call and put outcomes under the selected parameters. In the **hedging** module, the put reduces payoff variance and sharply limits severe losses, but the simulated probability of a negative payoff increases from 33.45% to 36.50% and the strategy requires a 0.57 premium. Accordingly, under the decision criterion used in Q3, the analysis supports **No Hedging**. The result highlights an important simulation insight: a hedge may improve some dimensions of risk while failing the particular measure used to make the final decision. Convergence diagnostics further show how running estimates stabilize as simulation size grows. Taken together, **the analysis demonstrates how Monte Carlo methods can move from modeling uncertain system evolution to evaluating financial values, risks, and explicit decisions**.

Selected References

Boyle, P., Broadie, M., & Glasserman, P. (1997). Monte Carlo methods for security pricing. Journal of Economic Dynamics and Control, 21, 1267–1321.

Glasserman, P. (2003). Monte Carlo Methods in Financial Engineering. Springer.

Kahale, N. (2018). General multilevel Monte Carlo methods for pricing discretely monitored Asian options. arXiv:1805.09427.

MIT Sloan School of Management. Analytics of Finance: Stochastic calculus, option pricing, and simulation methods. MIT OpenCourseWare.

Appendix A. Workbook Analytical Flow

Asian option: define parameters → generate weekly stock-price paths → compute path-average price → compute discounted call/put payoff → repeat 1,000 paths → estimate expected payoff, confidence intervals, threshold probabilities, histograms, and convergence.

Hedging: define parameters and put premium → simulate six-month terminal stock price → compute discounted put payoff → compute stock-only and stock-plus-put payoffs → repeat 2,000 scenarios → compare mean, variance, downside probability, extrema, empirical distributions, and convergence.